

Research Plan — Technical Report

Taxing liquidity on a decentralized prediction market: incidence, participation response, and who captures the maker subsidy

Draft technical report laying the foundations for one or more Q1 submissions in DeFi / Information Systems venues (e.g., *Information Systems Research*, *Journal of Management Information Systems*, *Decision Support Systems*, *Electronic Commerce Research and Applications*; finance-crossover: *Journal of Financial Markets*, *Management Science*).

1. Introduction

1.1 State of the art

Polymarket is the largest decentralized prediction market — an on-chain venue where users buy and sell binary outcome shares that settle to \$1 (if the event occurs) or \$0 (otherwise), so a share's price is a market-implied probability. Trading runs on a central limit order book (CLOB): a *maker* posts a resting limit order that provides liquidity, and a *taker* crosses the spread with a marketable order that consumes it. Settlement, custody, and order matching are recorded on the Polygon blockchain, so — unlike centralized exchanges — the near-complete trade history is publicly observable.

A cluster of 2025–2026 studies has established the market's first-order facts:

- **Who wins and loses.** Akey et al. (2026, SSRN 6443103), using the reconciled full history (~2.4M users, ~\$67B volume, 2022–2026), find that a small minority captures nearly all gains (top 1% of positive-PnL users $\approx$ 76.5% of profits; $\approx$ 69% of users end negative) and that liquidity provision (posting limit/maker orders) is the single strongest predictor of finishing profitable ($\approx$ +9 percentage points per standard-deviation increase in maker-volume share).
- **Skill vs. luck.** Gómez-Cram et al. (2026, SSRN 6617059), on ~1.72M accounts, use a sign-randomization test (repeatedly shuffle each account's buy/sell directions and compare actual profit to the resulting null distribution) and conclude that ~3% of accounts are persistently skilled, that market-makers plus that skilled minority (<3.5% of accounts) capture

30% of all gains, and that "the majority does not produce accuracy; it funds it."

- **Fill-side behavior.** Nechepurenko (2026, arXiv 2605.11640) attempted a behavioral tiering of *maker-side* participants on the on-chain fill tape but, in the current version, retracts the tiers — retaining only a one-cluster null and concluding that public fill records cannot identify market-making without the order/quote lifecycle (a ~2.3-day sample; no profit, no account-type layer).
- **Informed / insider trading.** Mitts & Ofir (2026, SSRN 6426778) build a per-(wallet, market) suspicion score for informed trading, using *newly created wallets* as one signal, and flag a large set of suspicious positions.
- **Foundations.** The Polymarket-v1 public archive (Qin & Yang, 2026, arXiv 2606.04217) supplies a reconciled, ground-truth trade record; Rahman et al. (2025, arXiv 2510.15612) give a systematization of knowledge (SoK) for decentralized-prediction-market microstructure and enumerate open research gaps (insider timing, coordination, MEV, replay).

1.2 The gap

The literature above answers *who profits* and *whether skill exists*. It does not address the economic event that most directly shapes those outcomes going forward: Polymarket's introduction of a trading fee. Specifically, three questions are untouched by all of the work in §1.1:

1. **Incidence and behavior of the fee.** No study measures the fee's *incidence* (who bears it, as a share of the value they trade), whether it is *regressive* (falls proportionally more on small traders), or how participants change behavior (order size, entry, exit) after it is introduced. The financial-economics literature has strong analogues — Umlauf (1993) and Colliard & Hoffmann (2017) on securities-transaction taxes, and the SEC tick-size pilot — but none on a prediction market, and only Whelan (working paper) models prediction-market fees theoretically.
2. **Who captures the maker subsidy.** Polymarket's fee is levied on takers and (partly) rebated to makers. The concurrent papers show *makers win on average* but not which makers capture the rebate, nor the mechanism. Adverse selection — the tendency for a passive liquidity provider to be "picked off" by better-informed counterparties (Glosten & Milgrom, 1985) — is the natural candidate mechanism and has never been measured on this venue.
3. **Who, structurally, is trading.** Prior work treats wallets as homogeneous users. It never resolves each wallet's on-chain account type — an externally owned account (EOA, a raw private key) versus a smart-contract wallet, and among the latter which *proxy* implementation (Polymarket's own factory proxy vs. a Gnosis Safe) — nor crosses that structural layer with a discrete behavioral-actor classification and profitability.

1.3 Why this matters for Polymarket (and DePMs broadly)

A prediction market is only socially useful if its prices aggregate information — which requires participation, and especially participation by the small, price-taking "crowd" whose flow the informed minority trades against. A fee that is regressive and that transfers value from the retail crowd to a professional market-making elite could erode exactly the participation that makes prices informative, while concentrating rents.

Whether that is happening — and who ends up bearing versus capturing the fee — is a first-order market-design and welfare question for Polymarket specifically and for the design of decentralized prediction markets generally. Because the entire ledger is on-chain, it is also a rare setting where these distributional questions can be answered on the *near-complete population* rather than a sample.

1.4 Research questions

We pose one main question, decomposed into four sub-questions.

Main RQ — How to measure, from public on-chain data, the incidence and distributional consequences of introducing a transaction fee on a decentralized prediction market?

Each sub-question isolates one facet of that measurement. For each, we state the question, why it is worth answering, and the gap in the literature it closes.

Sub-RQ 1 — What is the incidence of the fee across order sizes and trader roles, and is it regressive? *Why, and the gap it closes.* Incidence — who actually bears a levy, as a share of the value they trade — is the first-order welfare question about any transaction cost, and it has never been measured for a prediction market. The equity-market transaction-tax literature (Umlauf, 1993; Colliard & Hoffmann, 2017) measures aggregate liquidity and volume effects but not per-trader incidence, because it lacks the trader-level, whole-population record that an on-chain venue provides. This answer establishes whether the fee is flat or regressive — the fact the rest builds on.

Sub-RQ 2 — What behavioral and participation responses (order size, new-entrant inflow, churn) does the fee's introduction induce? *Why, and the gap it closes.* A prediction market's social value is its price accuracy, which depends on broad participation. If a fee shrinks orders and deters new entrants it erodes the very crowd whose flow makes prices informative. Prior prediction-market work (Whelan, working paper) models fee effects only theoretically; none observes the response to an actual fee introduction. The staggered, per-category rollout is a natural experiment that lets us observe it directly.

Sub-RQ 3 — What distinguishes the liquidity providers who capture the maker subsidy from those who lose it, and what role does adverse selection play? *Why, and the gap it closes.* The fee is levied on takers and partly rebated to makers, so its ultimate beneficiaries are whichever makers keep that rebate net of trading losses. Concurrent work (Akey et al., 2026; Gómez-Cram et al., 2026) shows makers win *on average* but not *which* makers, nor the mechanism. Adverse selection — a passive maker being "picked off" by better-informed counterparties (Glosten & Milgrom, 1985) — is the classic candidate mechanism and has never been measured on this venue. This answer names the beneficiary of the transfer and explains why.

Sub-RQ 4 — What is the on-chain account-type and behavioral-actor composition of the trader population, and how does account type map onto who pays versus who captures? *Why, and the gap it closes.* All prior work treats wallets as homogeneous users. None resolves each wallet's on-chain account type — an externally owned account (a raw key) versus a smart-contract wallet, and among the latter which proxy implementation — nor crosses that structural layer with a behavioral classification and profitability. This puts a concrete, structural identity on the payer and the beneficiary of Sub-RQs 1–3.

Together, the four answers compose the main question. Sub-RQ 1 fixes *what* is charged and *on whom* (the incidence); Sub-RQ 2 fixes *how the market responds* to that charge (the behavioral and welfare consequence); Sub-RQ 3 follows the money to *who ultimately keeps it* and *why* (the mechanism); Sub-RQ 4 attaches a *structural identity* to the payer and the beneficiary. Incidence, plus response, plus capture, plus identity is precisely what "the incidence and distributional consequences of introducing the fee, measured from public on-chain data" means — so the four sub-answers are jointly sufficient for the main RQ.

1.5 How we answer each (preview)

- Sub-RQ 1. The staggered per-category fee rollout is used as a natural experiment; fee incidence is computed per taker order and per order-size decile. Result: the fee is steeply regressive — small orders bear multiples of the rate that large orders do (§5.1).
- Sub-RQ 2. A balanced event study (constant set of categories around each category's own fee-start) measures order size and new-wallet inflow, netting out pre-trends. Result: no participation/volume response is identified — order size follows a pre-existing downtrend and new entry does not fall at fee onset; the earlier "orders halve / entry collapses" figures were panel-composition artifacts (§5.2).
- Sub-RQ 3. Maker-side realised PnL is decomposed against a counterparty-informedness proxy. Result: most makers lose; a professional/automated elite that escapes adverse selection captures the rebate (§5.3).
- Sub-RQ 4. Every trader is classified by on-chain contract type (from bytecode/storage) and by behavioral function (from activity timing), then crossed with PnL. Result: a ~97%-proxy population with an onboarding-channel split, in which market-makers and high-frequency automated accounts capture the edge (§5.4) — the economic direction replicates Akey/Gómez-Cram; the *account-type axis* is new to this literature.

2. Methodology

We describe methods in plain terms; technical terms are explained at first use.

2.1 Realised profit-and-loss (PnL). For each fill we compute the trader's realised PnL from the resolved outcome: buying a share of the winning outcome at price p for s shares yields $(1 - p) \cdot s$; buying the loser yields $-p \cdot s$ (symmetrically for sells). Summing over a wallet's fills gives its realised PnL. This is *outcome-dependent* — it uses the eventual result, so it measures realised (not risk-adjusted or forward-looking) performance and is subject to survivorship (wallets that stopped trading are still counted only where they appear). We report it as descriptive, never as certified skill.

2.2 Maker vs. taker and the fee. On each on-chain `OrderFilled` event we identify the *maker* (resting-order owner) and *taker* (aggressor). The fee (in the platform's 6-decimal USD unit) is charged once per taker order and recorded on the taker-order record; we therefore read fee incidence from those records and PnL from the maker-fill records, a distinction established and validated in the project audit (`docs/audit_2026-09-05.md`).

2.3 The fee rollout as a natural experiment. Fees were introduced staggered by market category over early 2026. We *recover* each category's fee-start month from the data (the first month its effective fee rate jumps above a threshold) and run an event study — outcomes aligned to "months relative to that category's own fee-start," with not-yet-treated categories serving as the implicit control (a difference-in-differences, DiD, design: comparing the change in treated units to the change in not-yet-treated units).

2.4 Fee incidence and regressivity. Incidence = fee paid ÷ notional traded, reported per order-size decile. "Regressive" means the rate is *higher* for smaller orders.

2.5 Adverse selection. A passive maker is *adversely selected* when it disproportionately trades against counterparties who turn out to be right. We proxy each maker's exposure by the size-weighted rate at which its taker counterparties' directional bets won, and relate it to the maker's PnL by Spearman rank correlation (a correlation robust to non-linearity and outliers).

2.6 Skill test (supporting). To separate skill from luck we use a sign-randomization (permutation) test: holding each wallet's positions, prices and outcomes fixed, we randomly flip the buy/sell direction of each *market position* many times; if the wallet's actual PnL is not better than this null, its side-selection is indistinguishable from chance. We report the fraction "certifiably skilled" both uncorrected and after Benjamini–Hochberg false-discovery-rate control (which bounds the expected share of false positives among the wallets we call skilled), and we cross-check with a correlation-immune split-half persistence test (does period-1 PnL predict period-2 PnL). We flag that this test's independence assumption is inflated by cross-market outcome correlation, so we treat the skilled fraction as an upper bound and the persistence result as the robust signal.

2.7 On-chain account-type classification. For a wallet, we call the Polygon node's `eth_getCode` to distinguish an EOA (externally owned account — no contract code, i.e. a raw key) from a smart-contract wallet. Contract wallets are keyed by their code hash and, for proxies, their implementation: an EIP-1167 minimal proxy embeds its implementation address in the bytecode; a Gnosis Safe proxy stores its singleton in a storage slot. This maps to Polymarket's documented `signature_type` (0 = EOA, 1 = Polymarket factory proxy for email/"Magic" users, 2 = Gnosis Safe for browser-wallet users). Classification is computed on an activity- and PnL-stratified sample plus the full set of top winners, with rate-limit-resilient, cached calls.

2.8 Behavioral-actor classification. Independently of the account type, we classify each active wallet (≥ 20 fills) by *function* using activity features: maker-share (fraction of fills as maker), fills per active day (intensity), and hour-of-day entropy (how evenly its trades spread across the 24 hours — a high value indicates near-24/7 activity consistent with automation). Heuristic labels: *market-maker* (maker-share ≥ 0.7), *hft_bot* (≥ 40 fills/active-day and near-24/7), *active_trader*, *retail_casual*, *one_shot*. Caveat (per Nechepurenko, 2026): public fills without the quote lifecycle cannot *prove* market-making; we therefore present these as heuristic behavioral classes, not identified market-maker status.

2.9 Entity / coordination (supporting, negative). We attempt to collapse wallets to beneficial owners via a shared-funding graph (wallets funded by a common non-hub USDC sender), excluding high-degree "hub" funders (exchange/relayer/platform

contracts), and we build a co-trading graph (same market, same side, same day). These test for sybil concentration and coordinated syndicates.

2.10 Memory-safe, reproducible computation. All population-scale aggregations are computed with DuckDB over the parquet store using wallet-hash / block-range bucketing (each aggregation processes a bounded fraction of the data with on-disk spill), so the full ~1.5-billion-row scans run within a fixed memory budget. Every figure is written to a versioned JSON artifact.

3. Data

3.1 Two eras, near-complete. We use two independent, near-complete on-chain records:

- **v1 archive** (Qin & Yang, 2026, arXiv 2606.04217; CC-BY): the reconciled Polymarket history, ~746 million maker-fill records across ~851k markets and ~2.6 million wallets, Nov 2022 – Apr 2026, with ground-truth aggressor direction, prices, per-fill fee, category, and resolved outcome. This is the *fee-free-to-fee-introduction* era (fees phased in during early 2026).
- **v2 on-chain tape:** our own extraction of the `OrderFilled` / `OrdersMatched` / `FeeCharged` events of the two v2 exchange contracts on Polygon, ~715 million raw / ~697 million deduplicated fills, Apr–Aug 2026 — the *mature-fee* era. Winner outcomes come from a verified on-chain conditional-tokens resolution registry (2,108,903 resolved conditions; token → winner derivation validated for both standard USDC-collateral and negRisk wrapped-collateral markets).

3.2 Coverage and comparability. Both eras cover essentially the whole population, matching the coverage of the concurrent full-population studies (Akey ≈2.4M users; Gómez-Cram ≈1.72M). The differentiator of this work is therefore not data coverage but the fee lens and the actor/ account-type classification applied to that population.

3.3 Known data caveats (carried into every result).

- **Taker-side tail (v1 feed only).** Some legacy per-wallet statistics derive from a taker-side 4,000-fill API tail; all population results here instead use the complete archive/tape, removing that bias. The earlier "no skilled cohort" finding was an artifact of that tail and is corrected.
 - **Chunk overlap (v2).** The resumable crawl can duplicate a fill across overlapping files; all v2 aggregates deduplicate on (tx-hash, event-index). Undeduplicated PnL was ~6–8% inflated; corrected.
 - **Fee location.** The taker fee lives on `is_taker_order = TRUE` records, not the maker-fill rows; net-of-fee PnL uses the former (audit §1, §8).
 - **Survivorship / outcome-dependence** (§2.1) applies throughout; results are descriptive.
 - **Fees vs. eras.** v1 is ~fee-free (gross ≈ net); v2 carries a ~1.17% effective taker fee, so cross-era PnL comparisons use v2 net of fees.
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4. State-of-the-art taxonomy

This section maps the recent literature against the analytical dimensions this work touches, dimension by dimension, so the gap the research questions address is visible directly in the table.

4.1 Coverage matrix

Work (year, id)	Population-scale	Who-wins / concentration	Skill test	Maker vs. taker	Fee incidence / behavior	Maker-subsidy capture / adverse selection	On-chain account TYPE	Behavioral-actor taxonomy
Akey et al. (2026, SSRN 6443103)	✓ full	✓ (top-1% ≈76.5%)	mixture model	✓ maker-share = top predictor (continuous)	✗	partial (maker-share → profit, no AS mechanism)	✗	✗ (continuous features, no discrete classes)
Gómez-Cram et al. (2026, SSRN 6617059)	✓ full	✓	✓ sign-randomization (~3% skilled)	✓ MMs separated functionally	✗	partial (MMs+skilled capture >30%)	✗	partial (MM vs skilled, skill-based)
Nechepurenko (2026, arXiv 2605.11640)	✗ (~2.3-day)	✗	✗	maker-side only	✗	✗	✗	attempted, then retracted (fills can't identify MM)
Mitts & Ofir (2026, SSRN 6426778)	large	✗	✗ (informed-trading score)	✗	✗	✗	wallet <i>provenance</i> (new wallets), not TYPE	✗
Whelan (working paper)	theory	✗	✗	✓ maker/taker economics	✓ theory only (no on-chain incidence)	partial (theory)	✗	✗
Qin & Yang (2026, arXiv 2606.04217)	✓ (dataset)	✗	✗	✓ (aggressor direction)	✗	✗	✗	✗
Rahman et al. (2025, arXiv 2510.15612)	SoK	—	—	—	lists as open	lists as open	✗	✗
This work	✓ full (both eras)	✓ (replicates)	✓ (reconciles)	✓	✓ empirical, staggered DiD	✓ AS mechanism, −0.58; who captures	✓ EOA/Safe/proxy/custom	✓ MM/HFT/active/retail (heuristic)

(✓ = addressed; partial = touched but not the focus; ✗ = not addressed.)

4.2 What each did and did not do

- The full-population papers (Akey; Gómez-Cram) own who-wins, concentration, skill, and the *maker-advantage direction*; they do not study the fee, the adverse-selection mechanism, the on-chain account type, or a discrete behavioral-actor taxonomy.
- Nechepurenko is the only prior fill-side behavioral attempt and it disclaims its own tiers on identifiability grounds — a direct methodological warning we adopt, and a gap we partly fill by adding profit and the whole population (while conceding the same identification limit for MM/HFT).
- Whelan supplies the theoretical fee framing but no on-chain incidence.
- Mitts & Ofir use wallet *provenance* (age) for insider detection — adjacent to, but not, an account-type taxonomy.

4.3 The gap the RQs address

Two columns of the matrix are empty across the entire literature: *fee incidence / behavior* and *maker-subsidy capture / adverse selection*; a third, *on-chain account type*, is empty as a research taxonomy (it exists only as documented product infrastructure — Polymarket's `signature_type`). These three empty columns are precisely Sub-RQs 1–2 (fee), 3 (subsidy / adverse selection), and 4 (account type). Conversely, the *who-wins / concentration / skill / maker-vs-taker* columns are filled — so we treat those results as a replication backdrop, cited to Akey and Gómez-Cram, never as our contribution. The lead contribution is therefore the fee experiment and the maker-subsidy mechanism, with the account-type axis as the novel supporting layer.

5. Results

All figures are computed on the full datasets described in §3.

5.1 A steeply regressive fee: the smallest orders bear the highest rates

Reading incidence per taker-order-size decile in the mature (v2) regime, the effective fee declines from ~2.7–3.2% of notional on the smallest orders to ~1.0% on the largest (`fee_behavioral.json`) — a broadly declining gradient (deciles 1–4 sit at 2.7–3.2%, decile 10 at 1.0%), not strictly monotone through the middle deciles. The archive rollout's per-order incidence shows the same steep decline, from an implausible ~514% on the smallest sub-dollar orders (a fixed-minimum-fee artifact on dust — we read the *gradient*, never these levels) down to ~8.8% on the largest. In the staggered rollout, on a balanced set of categories each category's effective fee jumps from ~0% pre-start to ~2–3% in its fee-start month and ~8% a month later (`fee_rollout_did.json`, `event_study_balanced`); a higher ~12% tail reflects one early-treated category, not the average. Over the mature-fee window (Apr–Aug 2026) the platform collected ~\$149.5M in taker fees (`v2_platform.json`). The fee is thus real, sharp at introduction, and regressive — it taxes small, price-taking flow proportionally hardest. (*Sub-RQ 1.*)

5.2 No participation response is identified once pre-trends are removed

A balanced event study — the six market categories present at every month of a $[-3, +1]$ window around each category's own fee-start, compared at constant composition (`fee_rollout_did.json`, `event_study_balanced`) — finds no participation response attributable to the fee. Median taker order size was **already falling before any fee** (pre-fee slope $\approx -\$0.51/\text{month}$); across the window the per-category mean runs $5.68 \rightarrow 5.96 \rightarrow 4.42 \rightarrow 3.85 \rightarrow 3.61$, and the decline at fee onset does not exceed that pre-trend ($-\$0.57$ from month -1 to 0 against a $-\$0.51$ trend) and in fact *decelerates* as the fee ramps to ~8% ($-\$0.24$ from 0 to $+1$). New-entrant inflow was **rising** pre-fee ($\approx +4,400$ wallets per category per month) and does not fall at onset — per category it runs $32,664 (-1) \rightarrow 46,542 (0) \rightarrow 37,744 (+1)$. The earlier pooled figures (order size $\sim \$6.9 \rightarrow \2.9 , new wallets $317\text{k} \rightarrow 89\text{k}$) were artifacts: that pooled series sums over a category set that shrinks from nine to one as the window widens (the $+2/+3$ tail is a single high-frequency category, crypto up/down) and does not net out the pre-existing downtrend. On the

balanced panel the volume/participation response is therefore **not identified**, consistent with the near-null elasticity expected for a small transaction cost. What is robust is the fee's sharp onset itself (effective rate $0 \rightarrow \sim 2\%$ at fee-start $\rightarrow \sim 8\%$ by +1). The welfare case for the fee thus rests on its **incidence and distribution** (§5.1, §5.3), not on a participation collapse. (*Sub-RQ 2.*)

5.3 The fee is a transfer to a professional market-making elite that escapes adverse selection

On the maker side of the archive, 637,314 of 1,214,275 maker wallets (52.5%) are net-negative (`maker_economics.json`) — just over half of all liquidity providers *lose* — and positive maker PnL is hyper-concentrated (Gini ≈ 0.95 ; the top 1% of winning makers take $\sim 69\%$). Yet makers as a class net +\$86.0M, the exact mirror of the takers' −\$86.1M. The split is driven by adverse selection: across makers, the size-weighted rate at which their taker counterparties' bets won is strongly negatively correlated with maker PnL (Spearman ≈ -0.58 , $p \approx 0$). Small/amateur makers (bottom volume deciles) face counterparties who win ~ 55 – 56% of the time and lose on average; the top-volume decile faces winning counterparties only $\sim 43\%$ of the time, earns $\sim +\$1,061$ mean, and captures +\$128.8M in aggregate — more than the entire net maker profit. Maker profitability is **non-monotonic in trading volume, with an interior minimum at the upper-middle (7th–9th) volume deciles**: these sizable but non-elite makers carry essentially all the aggregate losses (−\$24.4M, −\$12.0M, −\$9.9M) and face the most informed counterparties (win-rates 0.51, 0.54, 0.60), while only the top decile escapes adverse selection (0.43) and is net profitable. Size-weighted across *all* makers the counterparty win-rate is 0.43 (< 0.5): weighted by the volume that matters, the professional elite escapes adverse selection and the amateurs absorb it. Combined with §5.1–5.2 this frames the fee as a regressive transfer — small directional takers pay it; it is rebated to makers but captured by the professional, adverse-selection-avoiding minority, not the amateur LPs. The *direction* (liquidity provision captures the edge) replicates Akey and Gómez-Cram; the adverse-selection mechanism and the within-maker heterogeneity are the new elements. (*Sub-RQ 3.*)

5.4 Who trades: an on-chain account map and the actors who capture the edge

Account type. Resolving on-chain contract types on an activity- and PnL-stratified classified sample ($n \approx 27,111$) shows the population is $\sim 97\%$ smart-contract proxy wallets, split by onboarding channel: $\sim 74.5\%$ Gnosis Safe (browser-wallet users, `signature_type` 2), $\sim 21\%$ Polymarket factory proxy (email/"Magic" users, `signature_type` 1), $\sim 3.3\%$ EOA, and a $\sim 1\%$ tail of minor-variant contracts (`account_types.json`). Winners are almost exclusively the two dominant standard proxy types; the minor-variant contracts are low-activity one-offs, and we do not claim they are or are not bots. Mean realised PnL is highest for the email/Magic proxy ($\sim \$65k$) — but this is driven by a few extreme winners: the median account of every type is a net loser (−\$1 to −\$12), so account type by itself does *not* separate winners from losers. It is an onboarding-channel/infrastructure axis, not a skill axis.

Behavioral actors. Classifying active wallets by function (`actor_taxonomy.json`, 1.52M wallets):

actor type	n	maker-share	fills/day	24-7 (entropy frac)	mean PnL	win-rate
market_maker	84,120	0.84	16	0.72	+\$1,991	47%
hft_bot (near-24/7)	50,441	0.22	109	0.96	+\$1,152	29%
active_trader (directional)	274,031	0.13	32	0.63	-\$312	21%
retail_casual	1,086,388	0.18	4	0.67	-\$89	32%
one_shot	23,961	0.11	13	0.31	-\$346	23%

As a class, market-makers (+\$167.5M aggregate) and high-frequency near-24/7 automated accounts (+\$58.1M) are the net winners — capturing the edge via spread capture, not forecasting — while directional and retail classes lose in aggregate (−\$85.4M active, −\$96.5M retail). Within *every* class the median account is a small net loser, so the class-level edge sits in each class's right tail (mean $\gg$ median). This is precisely the population that captures the maker subsidy in §5.3. Two qualifications apply: the economic *direction* — liquidity provision and automation earn the edge — replicates the concurrent papers, and, because public fills do not carry the order/quote lifecycle (Nechepurenko, 2026), these are functional behavioral classes rather than identified market-maker roles. The new-to-the-literature contribution is the on-chain account-type axis and its crossing with function and PnL — most EOAs behave as retail (~62%), with a minority (~6% of the HFT class, ~1.8× over-represented) automated, so account type maps to *onboarding channel*, not cleanly to bot-vs-human. (*Sub-RQ 4.*)

5.5 Supporting / replicated backdrop (cited, not claimed as novel)

For completeness and to situate §5.1–5.4: takers lose −\$86.1M in the ~fee-free v1 era; in the fee era they are gross-positive (+\$20.0M) but −\$129.5M net — the ~1.17% taker fee (~\$149.5M) is the entire loss channel and then some, so the loss mechanism *shifts* from adverse selection (v1) to the fee (v2). Winnings are hyper-concentrated (Gini $\approx$ 0.95; the top 1% of winning wallets take ~70% of all winnings) — replicating Akey. On skill, a full-population sign-randomization test yields ~4.3% certifiably skilled after Benjamini–Hochberg correction (7.1% uncorrected) — the same order as Gómez-Cram's ~3%, and an upper bound given cross-market outcome correlation; the correlation-immune signal is weak-but-real persistence (split-half Spearman $\approx$ 0.15, $p \approx 0$; period-1 top-quintile $\rightarrow$ +\$1,174 mean in period 2 vs −\$134 for the bottom). Skill is expressed as breadth — the skilled tier trades far more markets (mean ~314 vs ~113 neutral) at a much higher market-level win-rate (~85%), not one lucky bet; a stricter position-level test in the informative price band is by contrast null (observed skilled winners 123 < chance 257), reinforcing the upper-bound reading. Coordination is a negative result: funding-graph entity collapse is small (~4.3% of top-20k-winner diversity is sybil), dominated by a single 377-wallet cluster (+\$48M) with the remainder singletons — no broad coordinated syndicate, consistent with the project's prior coordination null. These support, but are not, the contribution.

6. Limitations and next steps

- **Causal scope.** On a balanced-panel event study with a fee-free pre-trend removed, **no participation/volume response to the fee is identified** (§5.2); the incidence and distribution results (§5.1, §5.3) do not depend on the causal design. A placebo / synthetic-control design (below) would further bound any residual response, but the current reading is a null, not a magnitude to be sharpened.
- **Behavioral classes.** Market-maker and HFT labels are functional classes inferred from activity; public fills do not carry the order/quote lifecycle (Nechepurenko, 2026), so they are behavioral types, not identified roles.
- **Descriptive PnL.** All profit-and-loss is realised and outcome-dependent (subject to survivorship); it measures realised performance, not risk-adjusted skill.
- **Next analyses.** (a) A formal difference-in-differences with placebo categories and a synthetic control for the participation response; (b) a spread / adverse-selection decomposition of maker PnL where the order lifecycle can be reconstructed; (c) proxy → owner resolution (Gnosis Safe owners / factory owner) to test entity-level concentration directly; (d) net-of-fee incidence by account type, closing the link between the fee and account structure.

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